# Daniel Layeghi

London, UK

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I build robotics and machine-learning systems from idea to production. At Automata, I joined as employee #8, built the robot software from scratch, covering motion planning through embedded control, and led its three-person robotics team. That software later ran on robots deployed in NHS pathology laboratories; Automata subsequently raised a $50M Series B. I later completed my PhD in CS at Edinburgh University and published in RL and robotics, then co-founded Z2 Labs and built and marketed Souida, a book discovery platform.

## Founder & Industry Experience

**Z2 Labs** — London
*Co-founder* — Jun 2025–Aug 2026

Released and marketed a natural-language book-discovery product.

- Owned Souida's data, machine learning, retrieval, backend and production infrastructure.
- Built book-data ingestion, LLM annotation and embedding pipelines that processed approximately 10M tokens/min across 1M documents; developed custom FAISS-based multi-field retrieval, ranking and reranking at 50 QPS.
- Shipped containerised APIs and workers through Terraform-managed releases with parity checks and rollback.
- Reached 2.27k monthly unique visitors on Souida's consumer web product.
- Launched Souida on the web and marketed it through X, Instagram and TikTok; built an automated X recommendation agent integrated with the product.

**Automata** — London
*Robotics Engineer* — 2017–2019

Early-stage robotics company; subsequently raised $135M across three funding rounds.

- Joined as employee #8 and owned the robot software stack as Automata grew from eight to 40 employees.
- Wrote motion planning, trajectory optimisation and forward/inverse kinematics in C++; wrote control, rigid-body dynamics and collision detection in C for constrained microcontrollers, all from scratch.
- Built and deployed software for production testing, calibration and factory validation; established automated testing, CI/CD and controlled releases for the robot software.
- Worked directly with customers and the product team on the robot's first commercial deployments. The software later ran on robots deployed in NHS pathology laboratories.
- Interviewed candidates, hired two robotics engineers and technically led the three-person robotics team.

## Research Experience

**University of Edinburgh** — Edinburgh
*Postdoctoral Researcher (Amazon-funded)* — Mar 2024–Apr 2025
*PhD Researcher (UKRI-funded)* — Dec 2019–Mar 2024

- *Optimal Control via Combined Inference and Numerical Optimization* (ICRA 2022, Code).
  Consensus between sampling-based inference and second-order trajectory optimisation. (MuJoCo/C++/OpenMP).
- *Learning Long-Horizon Robot Manipulation Skills via Privileged Action* (CoRL 2025).
  Privileged actions and relaxed contact dynamics to improve exploration in long-horizon reinforcement learning (PyTorch).
- *Neural Lyapunov and Optimal Control* (arXiv, Code).
  Learned time-varying HJB value and Lyapunov functions for optimal-control policy learning (PyTorch).
- *Amortising Trajectory Optimisation for Residual MPC via Implicit Contact Differentiation* (RA-L submission, Code).
  Making MuJoCo differentiable at scale and enabling differentiable learning (JAX).
- Teaching & supervision: TA'd the Advanced Robotics course and supervised three MSc/BSc students.

## Education

**PhD, Computer Science**, University of Edinburgh — 2024
**MEng, Mechanical Engineering, First Class Honours**, Queen Mary University of London — 2017

## Selected Technical Skills

**Robotics & control:** motion planning, trajectory optimisation, kinematics, rigid-body dynamics, collision detection, model-based control, embedded control, differentiable simulation.
**Machine learning:** supervised and contrastive learning, reinforcement learning, probabilistic inference, neural value and Lyapunov methods, embeddings, retrieval, LLM annotation.
**Software & systems:** C, C++, Python, JAX, PyTorch, MuJoCo/MJX, FAISS, Redis, Docker, Terraform, CI/CD, Linux.